

Signal

A signal is basically information that changes according to something

- Voltage changing with time
- Temperature changing with altitude
- Pixel brightness changing with position
- Sound Pressure changing with time

Mathematically, we represent that relationship using a function

$x(t)$

Signal Value
[Output]

Independent Variable
[Input]

Important Notation

x : Whole Function / Signal

$x(t)$: Signal Value at $t \in \mathbb{R}$ ^{Continuous}

$x[n]$: Signal Value at $t \in \mathbb{Z}$ ^{Discrete}

Sampling

Sampling is the process of converting continuous-time (CT) to discrete-time (DT) by taking samples from CT at specified ^{Sampling Period} time intervals, T_s .

Suppose we have a originally CT signal, $x(t)$ and we take one measurement (sample) every T_s seconds.

$$x_s[0] = x(0)$$

$$x_s[1] = x(1 T_s)$$

$$x_s[2] = x(2 T_s)$$

⋮

$$x_s[n] = x(n T_s)$$

} After T_s seconds

} After T_s seconds

Trivially the

$$\text{sampling freq. } f_s = \frac{1}{T_s}$$

Signal / Function Manipulation

Transformations

Scaling

[Vertical Stretch]

$$y(t) = \alpha x(t)$$

Add

[Superposition]

$$\begin{aligned} f(x) &= x(t) \\ + \quad g(x) &= y(t) \\ \hline &= x(t) + y(t) \end{aligned}$$

Time Shift

[Horizontal Translation]

Shift RIGHT by α
 $y(t) = x(t - \alpha)$

Shift LEFT by α
 $y(t) = x(t + \alpha)$

Time Dilation

[Horizontal Stretch]

$$y(t) = x(\alpha t)$$

Take Note:

$\alpha < 0$ means
reflection about the
y-axis happens

Example



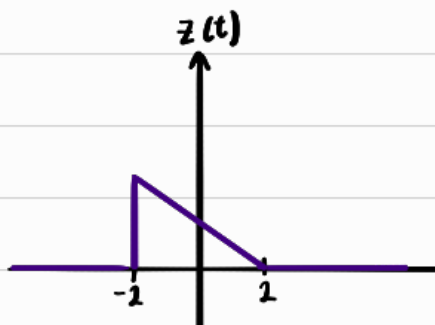
Draw the graph of y if ...
 $y(t) = x(-t + 2)$, $t \in \mathbb{R}$

Solution

Graph Given, $x(t)$

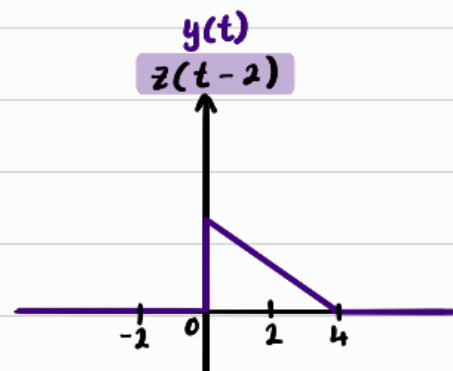
Reflect about
y-axis

$$\text{Let } z(t) = x(-t)$$



$$\begin{aligned} \text{Note that } z(t-2) &= x(-(t-2)) \\ &= x(-t+2) = y(t) \end{aligned}$$

Shift Right by 2

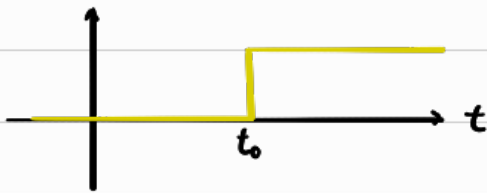


Unit step & Unit Impulse Signals

Unit Step Function

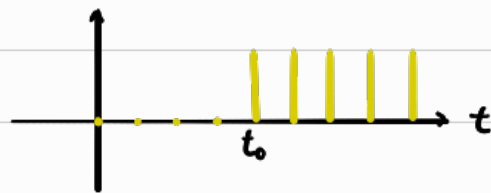
Continuous

$$u(t - t_0) = \begin{cases} 0 & \text{if } t < t_0 \\ 1 & \text{if } t > t_0 \end{cases}$$



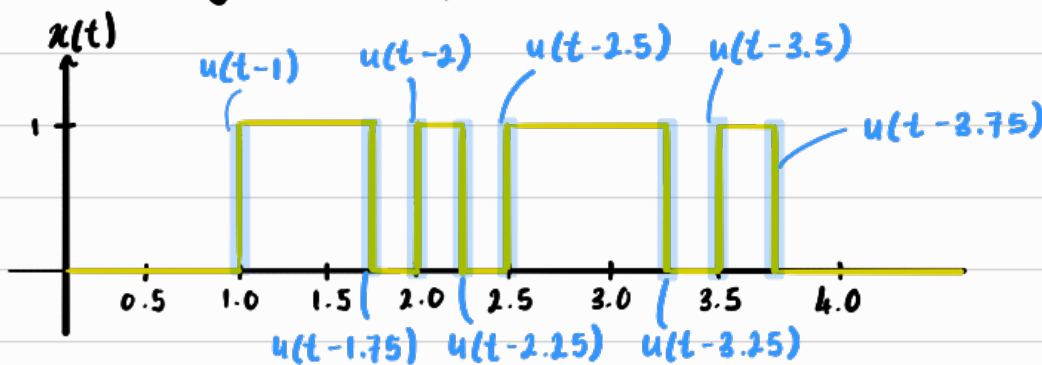
Discrete

$$u(n - n_0) = \begin{cases} 0 & \text{if } n < n_0 \\ 1 & \text{if } n \geq n_0 \end{cases}$$



Example

Express using unit step functions



Solution

Looking from Left to Right

$$\begin{aligned} x(t) = & [u(t-1) - u(t-1.75)] && \text{Pulse 1} \\ & + [u(t-2) - u(t-2.25)] && \text{Pulse 2} \\ & + [u(t-2.5) - u(t-3.25)] && \text{Pulse 3} \\ & + [u(t-3.5) - u(t-3.75)] && \text{Pulse 4} \end{aligned}$$

Unit Impulse [Delta Function]

Continuous [Dirac Delta]

$$\delta(t - t_0) = \begin{cases} \infty & \text{if } t = t_0 \\ 0 & \text{if } t \neq t_0 \end{cases}$$



Property : $\int_{-\infty}^{\infty} \delta(t) dt = 1$

(OR)

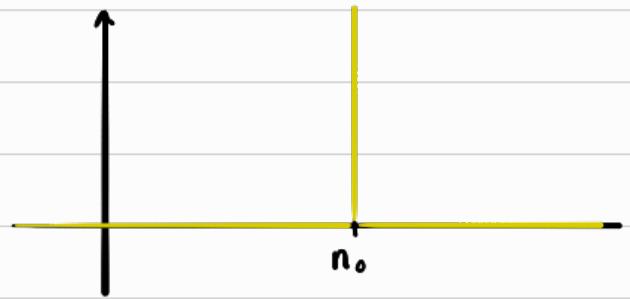
$$\int_{t_0 - \epsilon}^{t_0 + \epsilon} \delta(t - t_0) dt = 1$$

ϵ — Initially small number

$$\therefore \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$$

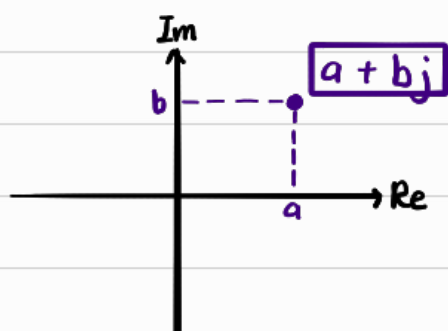
Discrete [Kronecker Delta]

$$\delta(n - n_0) = \begin{cases} 1 & \text{if } n = n_0 \\ 0 & \text{if } n \neq n_0 \end{cases}$$



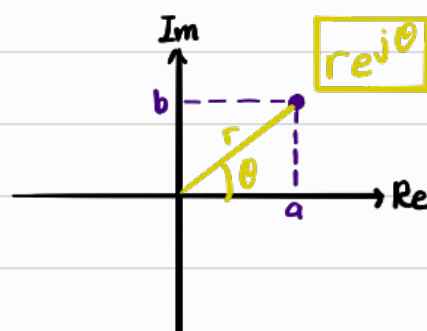
Complex Exponential Signals

Cartesian



$$x(t) = a + bj$$

Polar



$$x(t) = \underset{\text{Magnitude}}{r(t)} \underset{\text{Phase}}{e^{j\theta(t)}}, \text{ where } r(t) = |x(t)| = \sqrt{a^2 + b^2}$$

$[-\pi < \theta(t) \leq \pi]$

Unit complex exponentials [Continuous]

$$x(t) = \cos(\omega_0 t) + j \cdot \sin(\omega_0 t)$$

$$\theta = \omega_0 t \quad \downarrow \text{Using Euler, } e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$x(t) = \underset{\text{Magnitude}}{1} e^{j(\omega_0 t)} \quad \text{Rotation Speed}$$

General Complex Exponential [Continuous]

$$x(t) = e^{[\sigma_0 + j\omega_0]t} \implies x(t) = \underset{\text{Magnitude}}{e^{\sigma_0 t}} [e^{j\omega_0 t}]$$

Growth/Decay

Take Note : If $\sigma_0 = 0 \implies x(t) = e^{j(\omega_0 t)}$ [Unit Complex Exponential]

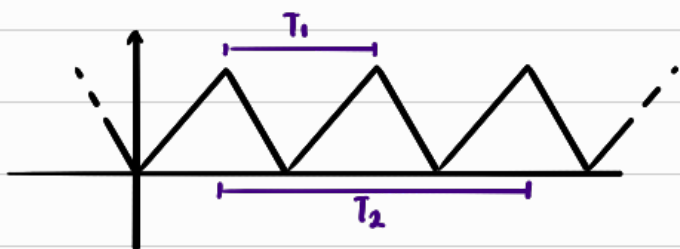
Take Note :

The discrete version of the Egn above are exactly the same, just that continuous time, $t \rightarrow$ Discrete value, n

Periodic Signal / Functions

Continuous Time (CT)

A CT signal is periodic if $x(t+T) = x(t)$, where $T = \text{period}$



As shown above, period, T isn't unique.
 T_1 & T_2 are valid periods

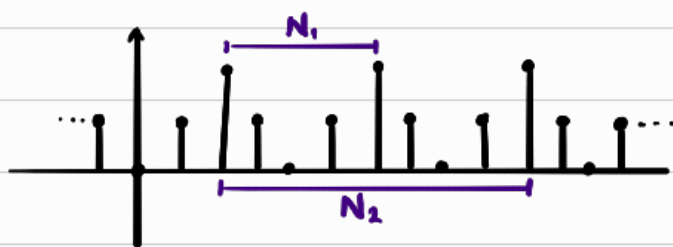
However the **fundamental period, T_0**
is the **smallest period, T** , thus
is unique.
 $\Rightarrow T_0 = T_1$

Fundamental Freq. (Hz), $f_0 = 1/T_0$

Fundamental Freq. (rad/s), $\omega_0 = 2\pi/T_0$
[Angular Frequency]

Discrete Time (DT)

A DT signal is periodic if $x[n+N] = x[n]$, where $N = \text{period}$



As shown above, period, N isn't unique
 N_1 & N_2 are valid periods

However the **fundamental period, N_0**
is the **smallest period, N** , thus
is unique.
 $\Rightarrow N_0 = N_1$

Fundamental Freq. (Hz), $f_0 = 1/N_0$

Fundamental Freq. (rad/sample), $\omega_0 = 2\pi/N_0$
[Angular Frequency]

Constant Signal Fundamental Period

For Example : $x(t) = 5$

Checking Periodicity, $x(t+T) = 5$

$\Rightarrow T$ can be any positive T

$\therefore$ There are infinitely many periods
 $\Rightarrow$ We don't have a **smallest period** **Fundamental Period**

For this course, it uses the convention

constant signal has frequency 0 [It never oscillates]

Periodicity of CT & DT

① CT unit complex exponential is always periodic

Consider $x(t) = e^{j\omega_0 t}$

We check $x(t+T) = x(t)$

$$\begin{aligned}x(t+T) &= e^{j\omega_0(t+T)} \\&= (e^{j\omega_0 t})(e^{j\omega_0 T}) \\&= x(t) \cdot e^{j\omega_0 T}\end{aligned}$$

$\therefore$ To satisfy $x(t+T) = x(t)$

$$\Rightarrow e^{j\omega_0 T} = 1$$

$$\omega_0 T = 1$$

$$= k(2\pi), k \in \mathbb{Z}$$

$$\therefore \boxed{T = \left(\frac{2\pi}{|\omega_0|}\right)k}$$

Absolute cuz ω_0 can < 0
but T can't

② DT complex exponential is NOT always periodic

Consider $x[n] = e^{j\omega_0 n}$

We check $x[n+N] = x[n]$

$$\begin{aligned}x[n+N] &= e^{j\omega_0(n+N)} \\&= (e^{j\omega_0 n})(e^{j\omega_0 N}) \\&= x[n] \cdot e^{j\omega_0 N}\end{aligned}$$

$\therefore$ To satisfy $x[n+N] = x[n]$

$$\Rightarrow e^{j\omega_0 N} = 1$$

$$\omega_0 N = m(2\pi), m \in \mathbb{Z}$$

$$\boxed{N = \left(\frac{2\pi}{\omega_0}\right)m}$$

Must be integer

If N isn't an integer, $x[n]$ is NOT periodic

Shortcut

Since N, m are integers,
 $\frac{2\pi}{\omega_0}$ must be a rational

Example

Find fundamental period of discrete-time signal x
 $x[n] = \sin\left(\frac{4\pi n}{5}\right)$ for all n

Compare with $\sin(\omega_0 n) \Rightarrow \omega_0 = \frac{4\pi}{5}$

Checking periodicity with formula

$$N = \left(\frac{2\pi}{\omega_0}\right)m$$

$$N = \left[\frac{2\pi}{4\pi/5}\right]m$$

$$= \frac{5}{2}m$$

$$\frac{N}{m} = \frac{5}{2}$$

$\therefore$ Choose smallest + integer N

$$m = 1 \Rightarrow N = 5/2$$

$$m = 2 \Rightarrow N = 5 \quad \text{[Fundamental Period]}$$

$$m = 3 \Rightarrow N = 15/2$$

$$m = 4 \Rightarrow N = 10$$

$\vdots$

$$\therefore N_0 = 5 \quad \#$$

Shortcut

$$\frac{N}{m} = \frac{\alpha}{\beta} \quad \text{Just make sure}$$

$\frac{\alpha}{\beta}$ is the simplest fraction

$$\Rightarrow N_0 = \alpha$$

Odd / Even Signals

Odd function [Anti-symmetrical]: $f(x) = -f(-x)$ [Reflect y-axis, then x-axis]

Even function [Symmetrical]: $f(x) = f(-x)$ [Reflect y-axis]

Easy Method to determine even/odd

Given $x(t)$

Step ① : Calculate $x(-t)$

Step ② : If $x(-t) = x(t) \Rightarrow$ even

If $x(-t) = -x(t) \Rightarrow$ odd

Bounded / Unbounded Signals

Bounded ~ Has a finite limit

Unbounded ~ No limit [Approaches Infinity]

Take note:

A complex exponential is bounded if the magnitude [dist of point from origin] never gets farther than some finite distance M from origin

Example

Is the following bounded or unbounded

$$x[n] = e^{j(\pi/2)n} + \cos(n)$$

$$x[n] = [e^{\cos(n)}] [e^{j(\pi/2)n}] \quad \begin{array}{l} \text{Unit Complex Exponential} \\ \text{[Magnitude = 1]} \end{array}$$

$$|x[n]| = e^{\cos(n)}$$

Since $\cos(n)$ is bounded btwn -1 and 1 ,

$\Rightarrow e^{\cos(n)}$ is also bounded

$\therefore x[n]$ is bounded $\neq$